# t3-w02 — condition C2

Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

> C1 plus the lesson note as source evidence.

## INPUT LAYERS

| Input layer | What it carries | Channel | In this condition |
|---|---|---|---|
| Scope | what is to be taught, and to whom | prompt | **present** |
| Prior knowledge | what the learner already knew coming in | prompt | **present** |
| Materials | what the lesson was represented with | prompt | **present** |
| Position | where this objective sat in the sequence | prompt | **present** |
| Method | the shape of the teaching, as event labels | prompt | **present** |
| Enacted content | what was actually explained and worked through | prompt | **present** |
| Set work | what the learners were asked to do in class | prompt | absent |
| Assessment | the questions the lesson was checked with | prompt | **present** |
| Figures | the drawings the lesson used | tool | **present** |

## WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

### Lesson context — shared by every section

**`subject`**

Mathematics

**`grade_level`**

JSS1

**`topic`**

Plane Shapes (2-D Shapes)

**`subtopic`**

Plane shapes: types, properties and perimeter

**`term`**

Third Term

**`week`**

2

**`lesson_objectives`**

- identify the types of plane shapes and state their properties, similarities and differences
- identify the parts of a circle
- find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

**entry_behaviour**

- Students have been taught angles.

**instructional_materials**

- Cardboard containing angles presentation.

### Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

**objective_text**

identify the types of plane shapes and state their properties, similarities and differences

**current_step**

1

**instructional_events**

- present_content
- elicit_performance
- present_content

**assessment_points**

- How many triangles has an octagon.

**lesson_content**


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Hints\n\nPlane shapes are also known as two dimensional shapes (2-D shapes).
They include: triangle,\nrectangle, squares, parallelogram, trapezium, circle and other polygons.
\n\n#### Triangles\n\nIsosceles: a single tick on each of the two slanted sides, an arc at each
base angle. Scalene:\none, two and three ticks on the three sides, an arc at each vertex.
Equilateral: a single tick\non each of the three sides, an arc at each vertex.\n\n[drawing
attached to this section here, unlabelled, marked Isosceles, Scalene, Equilateral]\n\n| Isosceles
| Scalene | Equilateral |\n|---|---|---|\n| (i) 2 adjacent side are equal | No side equal | 3
sides are equal |\n| (ii) base angles are equal | No angle is equal to the other | All angles are
equal ($60^\\circ$ each) |\n| (iii) one line of symmetry | No line of symmetry. | 3 lines of
symmetry. |\n\n#### Squares and Rectangles\n\nSquare: a right-angle mark at each corner, a single
tick on each of the four sides. Rectangle: a\nright-angle mark at each corner, a single tick on
each of the two longer sides and a double tick\non each of the two shorter sides.\n\n[drawing
attached to this section here, unlabelled, marked Squares, Rectangles]\n\n| Squares | Rectangles
|\n|---|---|\n| i) All sides are equal | 2 opposite sides are equal |\n| (ii) 2 pairs of parallel
sides | 2 pair of parallel sides. |\n| (iii) All angles are equal ($90^\\circ$) each. | All
angles are equal ($90^\\circ$) each. |\n| (iv) Four lines of symmetry | Two lines of symmetry |
\n| (v) Diagonals bisect it at right angle. | Diagonals bisect it but not at right angle. |
\n\n#### Parallelograms and Trapezium\n\nParallelogram: a double arrow on each of the two
horizontal sides, a single arrow on each of the\ntwo slanted sides. Trapezium: a single arrow on
each of the two parallel sides, and one diagonal\ndrawn in.\n\n[drawing attached to this section
here, unlabelled, marked Parallelogram, Trapezium]\n\n| Parallelogram | Trapezium |\n|---|---|\n|
i) 2 pairs of opposite side are equal and parallel | Only a pair of sides are parallel |\n| (ii)
opposite angles are equal | opposite angles are not equal |\n| (iii) No line of symmetry | No
line of symmetry |\n| (iv) Diagonals divide it into equal triangles | Diagonal divide it into two
unequal triangles. |\n\n#### Polygons\n\nA polygon is a closed shape that has three or more
straight sides.\n\n| Name of Polygon | No of sides | No of triangle ($n-2$) |\n|---|---|---|\n|
Triangle | 3 | 1 |\n| Quadrilateral | 4 | 2 |\n| Pentagon | 5 | 3 |\n| Hexagon | 6 | 4 |\n|
Heptagon | 7 | 5 |\n| Octagon | 8 | 6 |\n| Nonagon | 9 | 7 |\n| Decagon | 10 | 8 |\n| Hendecagon
or undecagon | 11 | 9 |\n| Dodecagon | 12 | 10 |"
 }
]
```

**Withheld from this payload:** `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

---

### Objective 2 — identify the parts of a circle

**`objective_text`**

identify the parts of a circle

**`current_step`**

2

**`instructional_events`**

• present_content

**`lesson_content`**


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Parts of a circle\n\nTwo labelled circles. The first has the diameter drawn
across it and a radius from its centre to\nthe circumference, with the arc and the circumference
labelled. The second has a chord near the\ntop with the segment above it shaded, a shaded sector
below with its right-hand bounding line\nlabelled Radius, and a tangent touching it on the right.
\n\n[drawing attached to this section here, unlabelled, marked Diameter, Radius, arc,
Circumference, Chord, Segment, Sector, Tangent]\n\n- **Arc:** This is a curved part of the
circumference.\n- **Radius:** This is any straight line joining the centre of the circle to the
circumference.\n- **Chord:** This is any straight line joining two point on the circumference of
a circle.\n- **Diameter:** Any chord that passes through the centre of the circle."
 }
]
```


**Withheld from this payload:** `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

### Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

**objective_text**

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

**current_step**

3

**assessment_points**

• Find the perimeter of a square of sides 8cm
• Find the perimeter of a rectangle of length 7cm and breadth 4cm.
• What is the perimeter of regular pentagon with sides 5cm.
• Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
• The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
• A rectangle is such that its length is twice its breadth, if the breadth is $d$, find the total length of the sides of rectangle.
• Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
• Find the perimeter of a regular hexagon with sides 5cm.

**lesson_content**


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Perimeter\n\nPerimeter is the distance round a shape.\n\n**Worked
examples**\n\n##### 1. Calculate the perimeter of a square of side 15cm.\n\n[drawing attached to
this section here, unlabelled, marked Square, L 15cm, 15cm, L]\n\n**Solution**\n\nPerimeter of a
square $= L + L + L + L = 4L$\n\n$= 15 + 15 + 15 + 15 = 4(15)$\n\n$= 60\\mathrm{cm}$\n\n##### 2.
Find the perimeter of a rectangle whose length 10cm and breadth 7cm.\n\n[drawing attached to this
section here, unlabelled, marked L 10cm, L, b, 7cm]\n\n**Solution**\n\nPerimeter of rectangle $=
L + b + L + b$\n\n$= L + L + b + b$\n\n$= 2L + 2b$\n\n$= 2(L + B)$\n\n$= 10\\mathrm{cm} + 7\
\mathrm{cm} + 10\\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 10\\mathrm{cm} + 10\\mathrm{cm} + 7\
\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 2(10) + 2(7)$\n\n$= 2(10 + 7)$\n\n$= 2(17)$\n\n$= 34\
\mathrm{cm}$\n\n##### 3. If the perimeter of a rectangle is 60cm and its length is 17cm.
Calculate its breadth.\n\n[drawing attached to this section here, unlabelled, marked L = 17 cm,
b]\n\n**Solution**\n\nPerimeter $= 60\\mathrm{cm}$\n\n$P = L + b + L + b$\n\n$60 = 17\\mathrm{cm}
+ b + 17\\mathrm{cm} + b$\n\n$60 = 17\\mathrm{cm} + 17\\mathrm{cm} + b + b$\n\n$60\\mathrm{cm} =
2(17\\mathrm{cm}) + 2b$\n\n$60\\mathrm{cm} = 2(17\\mathrm{cm} + b)$\n\n$60\\mathrm{cm} = 34\
\mathrm{cm} + 2b$\n\ncollect like terms.\n\n$60\\mathrm{cm} - 34\\mathrm{cm} = 2b$\n\n$2b = 26\
\mathrm{cm}$\n\nDivide both side by 2\n\n$\\frac{2b}{2} = \\frac{26\\mathrm{cm}}{2}$\n\n$b = 13\
\mathrm{cm}$\n\n##### 4. Find the perimeter of the trapezium below.\n\n[drawing attached to this
section here, unlabelled, marked a 11cm, c 5cm, 9cm, b, 4cm, d]\n\nPerimeter of trapezium $= a +
b + c + d$\n\n$= 11\\mathrm{cm} + 9\\mathrm{cm} + 5\\mathrm{cm} + 4\\mathrm{cm}$\n\n$= 29\
\mathrm{cm}$\n\n##### 5. Find the perimeter of the parallelogram below.\n\n[drawing attached to
this section here, unlabelled, marked 9cm a, a, b, 5cm]\n\n**Solution**\n\nPerimeter of
parallelogram $= a + a + b + b$\n\n$= 2a + 2b$\n\n$= 2(a + b)$\n\n$= 2(9 + 5)$\n\n$= 2(14)$\n\n$=
28\\mathrm{cm}$\n\n##### 6. Find the perimeter of a regular hexagon sides 5.2cm.
\n\n**Solution.**\n\n[drawing attached to this section here, unlabelled, marked Regular hexagon,
5.2cm]\n\nHexagon has 6 sides.\n\nPerimeter $= 6 \\times 5.2\\mathrm{cm}$\n\n$= 31.2\\mathrm{cm}$
\n\n##### 7. Calculate the perimeter of circle of radius 7cm (use the value $3\\frac{1}{7}$ for $
\\pi$)\n\n**Solution**\n\n[drawing attached to this section here, unlabelled, marked 7cm]
\n\nPerimeter of a circle $= 2\\pi r$ or $\\pi d$\n\n$= 2 \\times 3\\frac{1}{7} \\times 7\
\mathrm{cm}$\n\n$= 2 \\times \\frac{22}{7} \\times 7$\n\n$= 2 \\times 22 = 44\\mathrm{cm}$
\n\nusing $\\pi d$.\n\nSince $r = 7\\mathrm{cm}$\n\n$d = 2 \\times r = 2 \\times 7\\mathrm{cm} =
14\\mathrm{cm}$\n\n$\\frac{22}{7} \\times 14\\mathrm{cm}$\n\n$= 22 \\times 2$\n\n$= 44\
\mathrm{cm}$\n\n##### 8. A rope is wound 50 times round a cylinder of radius 25cm. How long is
```

```
the rope? $\\pi = 3.14$.\n\n**Solution**\n\n[drawing attached to this section here, unlabelled]
\n\nlength of the rope $=$ circumference.\n\nCircumference $= 2\\pi r$\n\n$= 2 \\times 3.14 \
\times 25$\n\n$= 157\\mathrm{cm}$\n\nTotal length of the rope $= 157\\mathrm{cm} \\times 50$
\n\n$= 7850\\mathrm{cm}$.\n\n##### 9. The circumference of a circle is 440cm. Find the radius $(\
\pi = \\frac{22}{7})$\n\n**Solution**\n\n[drawing attached to this section here, unlabelled,
marked r]\n\nCircumference $= 2\\pi r$\n\n$440 = 2 \\times \\frac{22}{7} \\times r$\n\n$440 = \
\frac{44r}{7}$\n\ncross multiply.\n\n$440 \\times 7 = 44r$\n\n$3080 = 44r$\n\ndivide both side by
$44$\n\n$\\frac{44r}{44} = \\frac{3080}{44}$\n\n$r = 70\\mathrm{cm}$\n\n##### 10. Calculate the
perimeter of the shape below.\n\n[drawing attached to this section here, unlabelled, marked 4.9
cm, 4.9cm]\n\n**Solution**\n\nlength of the curved part $= \\frac{1}{4} \\times 2\\pi r$\n\n$= \
\frac{1}{4} \\times 2 \\times \\frac{22}{7} \\times 4.9$\n\n$= \\frac{1}{4} \\times \\frac{44}{7}
\\times 4.9$\n\n$= \\frac{11}{7} \\times \\frac{49}{10}$\n\n$= \\frac{11 \\times 7}{10}$\n\n$= \
\frac{77}{10}$\n\n$= 7.7\\mathrm{cm}$\n\nPerimeter of the shape $= 4.9 + 4.9 + 7.7$\n\n$= 17.5\
\mathrm{cm}$"
 }
]
```


**Withheld from this payload:** `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

## WHAT THE MODEL PRODUCED

### Identifying Plane Shapes and Their Properties

EXPLANATION

Plane shapes (2-D shapes) are figures that lie in a flat surface. They can be grouped by the number of sides they have and by the way those sides and angles relate to each other.

**Triangles** – three sides and three angles (fig-01).
- *Isosceles*: two sides equal, the angles opposite those sides are equal, one line of symmetry.
- *Scalene*: no sides equal, no angles equal, no line of symmetry.
- *Equilateral*: all three sides equal, all three angles are $60^{\circ}$, three lines of symmetry.

**Quadrilaterals** – four sides.
- *Square* (fig-02): four equal sides, four right angles, four lines of symmetry, diagonals bisect each other at right angles.
- *Rectangle* (fig-02): opposite sides equal, four right angles, two lines of symmetry, diagonals bisect each other but not at right angles.
- *Parallelogram* (fig-03): opposite sides equal and parallel, opposite angles equal, no lines of symmetry, diagonals cut the shape into two equal triangles.
- *Trapezium* (fig-03): only one pair of opposite sides parallel, no lines of symmetry, a diagonal divides it into two unequal triangles.

**Similarities and differences**
- All squares, rectangles, parallelograms and trapeziums are quadrilaterals, so they each have four sides.
- Squares and rectangles share four right angles, but only squares have all sides equal.
- Parallelograms and rectangles both have opposite sides parallel, yet only rectangles have right angles.
- Trapeziums differ by having just one pair of parallel sides.

Remember: the tick marks, arrows and arcs in the figures are symbols that tell you which sides or angles are equal or right-angled. Use those clues to recognise each shape quickly.

WORKED_EXAMPLE

**Worked example:** Identify the shape shown below.

The figure has four sides, all of the same length, and each corner has a right-angle mark. The tick marks on all four sides indicate they are equal. This matches the description of a **square** (fig-02).

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(4,0)--(2,4)--cycle;
  \draw (0.8,1.6)--(1.2,1.4);
  \draw (3.2,1.6)--(2.8,1.4);
  \draw (0,0)++(0:0.7) arc[start angle=0, end angle=63, radius=0.7];
  \draw (4,0)++(180:0.7) arc[start angle=180, end angle=117, radius=0.7];
  \node[below] at (2,-0.6) {Isosceles};
  \begin{scope}[xshift=7cm]
    \draw (0,0)--(5,0)--(1.4,3.4)--cycle;
    \draw (0.38,1.83)--(1.02,1.57);
    \draw (3.02,1.87)--(3.50,2.38);
    \draw (3.38,1.53)--(3.86,2.04);
    \foreach \d in {-0.25,0,0.25} { \draw (2.5+\d,-0.25)--(2.5+\d,0.25); }
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=68, radius=0.7];
    \draw (5,0)++(180:0.8) arc[start angle=180, end angle=137, radius=0.8];
    \draw (1.4,3.4)++(-113:0.7) arc[start angle=-113, end angle=-25, radius=0.7];
    \node[below] at (2.5,-0.6) {Scalene};
  \end{scope}
  \begin{scope}[xshift=14cm]
    \draw (0,0)--(4,0)--(2,3.46)--cycle;
    \draw (0.8,1.55)--(1.2,1.32);
    \draw (3.2,1.55)--(2.8,1.32);
    \draw (1.8,-0.25)--(2.2,0.25);
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=60, radius=0.7];
```

```
    \draw (4,0)++(180:0.7) arc[start angle=180, end angle=120, radius=0.7];
    \draw (2,3.46)++(-120:0.7) arc[start angle=-120, end angle=-60, radius=0.7];
    \node[below] at (2,-0.6) {Equilateral};
  \end{scope}
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (3.4,3.4);
  \draw (0,0.5)--(0.5,0.5)--(0.5,0);
  \draw (3.4,0.5)--(2.9,0.5)--(2.9,0);
  \draw (0,2.9)--(0.5,2.9)--(0.5,3.4);
  \draw (3.4,2.9)--(2.9,2.9)--(2.9,3.4);
  \draw (1.7,-0.25)--(1.7,0.25);
  \draw (1.7,3.15)--(1.7,3.65);
  \draw (-0.25,1.7)--(0.25,1.7);
  \draw (3.15,1.7)--(3.65,1.7);
  \node[below] at (1.7,-0.9) {Squares};
  \begin{scope}[xshift=8cm]
    \draw (0,0) rectangle (5,3);
    \draw (0,0.5)--(0.5,0.5)--(0.5,0);
    \draw (5,0.5)--(4.5,0.5)--(4.5,0);
    \draw (0,2.5)--(0.5,2.5)--(0.5,3);
    \draw (5,2.5)--(4.5,2.5)--(4.5,3);
    \draw (2.5,-0.25)--(2.5,0.25);
    \draw (2.5,2.75)--(2.5,3.25);
    \draw (-0.25,1.35)--(0.25,1.35); \draw (-0.25,1.65)--(0.25,1.65);
    \draw (4.75,1.35)--(5.25,1.35); \draw (4.75,1.65)--(5.25,1.65);
    \node[below] at (2.5,-0.9) {Rectangles};
  \end{scope}
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(5,0)--(6.6,3)--(1.6,3)--cycle;
  \draw[->] (2.4,0)--(2.6,0);
  \draw[->] (3.9,3)--(4.1,3);
  \draw[->] (5.7,1.4)--(5.85,1.65);
  \draw[->] (5.85,1.4)--(6.0,1.65);
  \draw[->] (0.7,1.4)--(0.85,1.65);
  \draw[->] (0.85,1.4)--(1.0,1.65);
  \node[below] at (3,-0.6) {Parallelogram};
  \begin{scope}[xshift=10cm]
    \draw (0,0)--(6,0)--(4.6,3)--(1.6,3)--cycle;
    \draw (0,0)--(4.6,3);
    \draw[->] (2.9,0)--(3.1,0);
    \draw[->] (3.0,3)--(3.2,3);
    \node[below] at (3,-0.6) {Trapezium};
  \end{scope}
\end{tikzpicture}
```

**1.** 1. Which triangle has three equal sides and three equal angles?
answer: Equilateral triangle

**2.** 2. Name the quadrilateral that has opposite sides equal and parallel but does not have right angles.
answer: Parallelogram

**3.** 3. How many lines of symmetry does a square have?
answer: 4

**4.** 4. Which shape has only one pair of parallel sides?
answer: Trapezium

**5.** 5. List two properties that squares and rectangles share.

answer: Both have four right angles and opposite sides parallel.

### Parts of a Circle

EXPLANATION

A circle is made up of several important parts:

- **Radius** – a straight line from the centre of the circle to any point on the circumference.
- **Diameter** – a special chord that passes through the centre; it is twice the length of a radius.
- **Chord** – any straight line joining two points on the circumference.
- **Arc** – a curved part of the circumference between two points.
- **Sector** – the region bounded by two radii and the arc between them.
- **Tangent** – a straight line that touches the circle at exactly one point.

These parts are shown in figure fig-04. Notice how the diameter cuts the circle into two equal halves, how the radius reaches out from the centre, and how the chord does not have to pass through the centre. The arc is the curved edge, while a sector looks like a ‘slice of pizza’. A tangent just grazes the circle without cutting it.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (-3,0)--(3,0);
  \node[above] at (0,0.1) {Diameter};
  \draw (0,0)--(0,-3);
  \node[right] at (0.45,-1.9) {Radius};
  \draw[->] (-5.9,-3.4)--(-2.3,-1.9);
  \node[left] at (-5.9,-3.4) {arc};
  \draw[->] (4.4,3.4)--(2.2,2.2);
  \node[right] at (4.4,3.4) {Circumference};
  \begin{scope}[xshift=17cm]
    \draw (0,0) circle (3);
    \draw (-2.29,1.94)--(2.29,1.94);
    \begin{scope}
      \clip (0,0) circle (3);
      \clip (-3,1.94) rectangle (3,3);
      \foreach \x in {-3,-2.7,...,3} { \draw (\x,1.8)--(\x+1.2,3.0); }
    \end{scope}
    \node[below] at (0,1.9) {Chord};
    \draw[->] (3.2,4.2)--(0.6,2.5);
    \node[right] at (3.2,4.2) {Segment};
    \begin{scope}
      \clip (0,0)--(-80:3) arc[start angle=-80, end angle=-25, radius=3]--cycle;
      \foreach \y in {-3.4,-3.1,...,0} { \draw (0,\y)--(3,\y+1.2); }
    \end{scope}
    \draw (0,0)--(-80:3);
    \draw (0,0)--(-25:3) node[midway, sloped, above] {Radius};
    \draw[->] (-3.6,-4.2)--(1.0,-2.0);
    \node[left] at (-3.6,-4.2) {Sector};
    \draw (3,-2.4)--(3,2.4);
    \node[right] at (3.2,0) {Tangent};
  \end{scope}
\end{tikzpicture}
```

WORKED_EXAMPLE

**Worked Example**

*Identify the parts shown in the diagram.*

In the first circle of fig-04:
- The line drawn across the centre is the **Diameter**.
- The line from the centre to the edge is a **Radius**.

> - The curved line around the edge is the **Circumference**, and a part of it is an **Arc**.
>
> In the second circle of fig-04:
> - The straight line joining two points on the edge is a **Chord**.
> - The shaded region bounded by two radii and the arc is a **Sector**.
> - The line that just touches the circle on the right is a **Tangent**.
>
> Remember: every diameter is a chord, but not every chord is a diameter.

**1.** 1. In a circle, a line from the centre to the edge is called a _____.
answer: Radius

**2.** 2. A chord that passes through the centre of the circle is called a _____.
answer: Diameter

**3.** 3. The part of the circumference between two points is known as a _____.
answer: Arc

**4.** 4. The region bounded by two radii and the arc between them is a _____.
answer: Sector

**5.** 5. A straight line that touches a circle at exactly one point is a _____.
answer: Tangent

### Finding Perimeters of Common Shapes

EXPLANATION
The perimeter of a shape is the total distance around it. For regular polygons and many other shapes you can add the lengths of all the sides.

- **Square**: four equal sides → $P = 4 \times \text{side}$.
- **Rectangle**: two lengths $L$ and two breadths $B$ → $P = 2(L + B)$.
- **Regular polygon** (all sides equal): $P = n \times \text{side}$ where $n$ is the number of sides.
- **Parallelogram**: opposite sides equal → $P = 2(a + b)$.
- **Trapezium**: add the four side lengths $P = a + b + c + d$.
- **Circle**: perimeter is called the circumference and is $C = 2\pi r = \pi d$. Use $\pi = \frac{22}{7}$ unless otherwise stated.

WORKED_EXAMPLE
**Example 1 – Square**
A square has each side 15 cm (see fig-05).

$P = 4 \times 15 = 60$ cm.

WORKED_EXAMPLE
**Example 2 – Rectangle**
Length $L = 10$ cm, breadth $B = 7$ cm (see fig-06).

$P = 2(L + B) = 2(10 + 7) = 34$ cm.

WORKED_EXAMPLE
**Example 3 – Regular Pentagon**
Each side is 5 cm. A pentagon has 5 sides.

$P = 5 \times 5 = 25$ cm.

WORKED_EXAMPLE
**Example 4 – Trapezium**
Sides: $a = 11$ cm, $b = 9$ cm, $c = 5$ cm, $d = 4$ cm (see fig-08).

$P = a + b + c + d = 11 + 9 + 5 + 4 = 29$ cm.

WORKED_EXAMPLE
**Example 5 – Parallelogram**
Sides $a = 9$ cm and $b = 5$ cm (see fig-09).

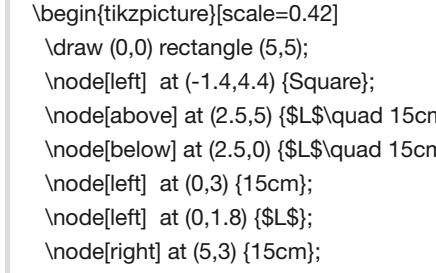
$P = 2(a + b) = 2(9 + 5) = 28$ cm.

WORKED_EXAMPLE
**Example 6 – Regular Hexagon**
Each side is 5.2 cm (see fig-10).

$P = 6 \times 5.2 = 31.2$ cm.

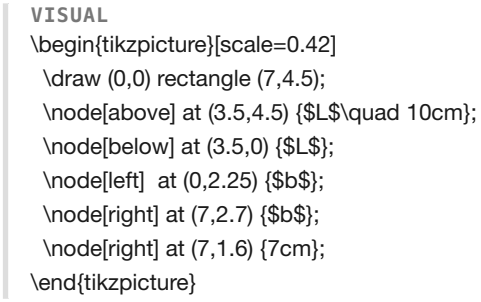
WORKED_EXAMPLE
**Example 7 – Circle**
Radius $r = 7$ cm, use $\pi = \frac{22}{7}$ (see fig-11).

$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm.

VISUAL


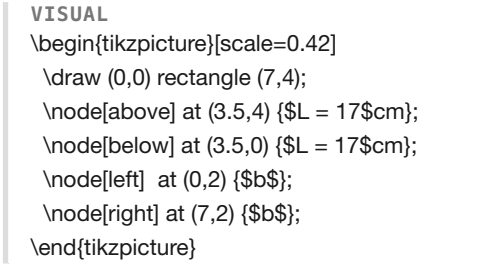
```
  \node[right] at (5,1.8) {$L$};
\end{tikzpicture}
```

VISUAL



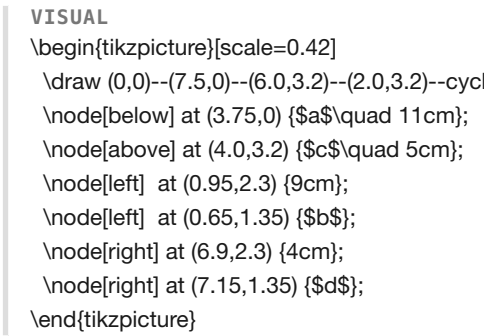
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4.5);
  \node[above] at (3.5,4.5) {$L$\quad 10cm};
  \node[below] at (3.5,0) {$L$};
  \node[left]  at (0,2.25) {$b$};
  \node[right] at (7,2.7) {$b$};
  \node[right] at (7,1.6) {7cm};
\end{tikzpicture}
```

VISUAL



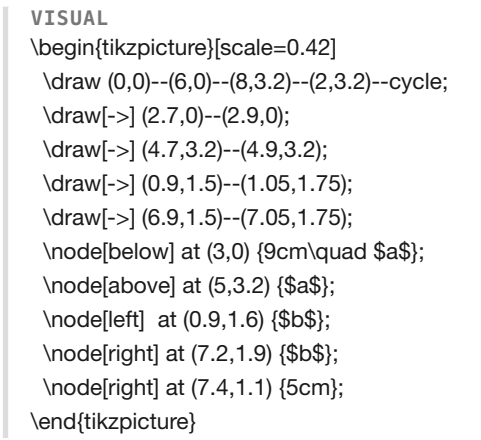
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4);
  \node[above] at (3.5,4) {$L = 17$cm};
  \node[below] at (3.5,0) {$L = 17$cm};
  \node[left]  at (0,2) {$b$};
  \node[right] at (7,2) {$b$};
\end{tikzpicture}
```

VISUAL



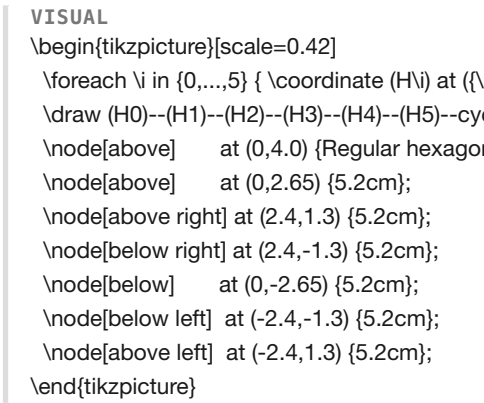
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(7.5,0)--(6.0,3.2)--(2.0,3.2)--cycle;
  \node[below] at (3.75,0) {$a$\quad 11cm};
  \node[above] at (4.0,3.2) {$c$\quad 5cm};
  \node[left]  at (0.95,2.3) {9cm};
  \node[left]  at (0.65,1.35) {$b$};
  \node[right] at (6.9,2.3) {4cm};
  \node[right] at (7.15,1.35) {$d$};
\end{tikzpicture}
```

VISUAL



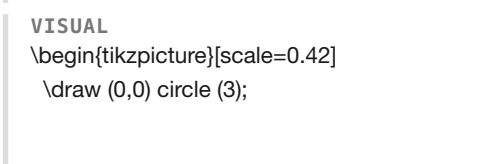
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(6,0)--(8,3.2)--(2,3.2)--cycle;
  \draw[->] (2.7,0)--(2.9,0);
  \draw[->] (4.7,3.2)--(4.9,3.2);
  \draw[->] (0.9,1.5)--(1.05,1.75);
  \draw[->] (6.9,1.5)--(7.05,1.75);
  \node[below] at (3,0) {9cm\quad $a$};
  \node[above] at (5,3.2) {$a$};
  \node[left]  at (0.9,1.6) {$b$};
  \node[right] at (7.2,1.9) {$b$};
  \node[right] at (7.4,1.1) {5cm};
\end{tikzpicture}
```

VISUAL



```
\begin{tikzpicture}[scale=0.42]
  \foreach \i in {0,...,5} { \coordinate (H\i) at ({\i*60}:3); }
  \draw (H0)--(H1)--(H2)--(H3)--(H4)--(H5)--cycle;
  \node[above]       at (0,4.0) {Regular hexagon};
  \node[above]       at (0,2.65) {5.2cm};
  \node[above right] at (2.4,1.3) {5.2cm};
  \node[below right] at (2.4,-1.3) {5.2cm};
  \node[below]       at (0,-2.65) {5.2cm};
  \node[below left]  at (-2.4,-1.3) {5.2cm};
  \node[above left]  at (-2.4,1.3) {5.2cm};
\end{tikzpicture}
```

VISUAL



```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
```

```
  \draw (0,0)--(3,0) node[midway,above] {7cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) ellipse (2 and 0.6);
  \draw (-2,0)--(-2,-6);
  \draw (2,0)--(2,-6);
  \draw (-2,-6) arc[start angle=180, end angle=360, x radius=2, y radius=0.6];
  \foreach \y in {-0.9,-1.8,-2.7,-3.6,-4.5,-5.4} {
    \draw (-2,\y) arc[start angle=180, end angle=360, x radius=2, y radius=0.55];
    \draw (2,\y-0.03)--(-2,\y-0.9);
  }
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {$r$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(0,5)--(0,5) arc[start angle=90, end angle=0, radius=5]--(0,0);
  \node[left]  at (0,2.5) {4.9 cm};
  \node[below] at (2.5,0) {4.9cm};
\end{tikzpicture}
```

**1.** Find the perimeter of a square whose side is 8 cm.
answer: 32 cm

**2.** Find the perimeter of a rectangle with length 7 cm and breadth 4 cm.
answer: 22 cm

**3.** What is the perimeter of a regular pentagon with each side 5 cm?
answer: 25 cm

**4.** Find the perimeter of a parallelogram whose base is 6 cm and side is 5 cm.
answer: 22 cm

**5.** The circumference of a circle is 22 cm. Find the radius (use \(\pi = \frac{22}{7}\)).
answer: 1 cm

**6.** A rectangle has length twice its breadth. If the breadth is \(d\) cm, write an expression for the total length of all four sides.
answer: 6d cm

**7.** Calculate the perimeter of the figure shown with side lengths 1 cm, 5 cm and 9 cm (see fig-14).
answer: 15 cm

**8.** Find the perimeter of a regular hexagon with each side 5 cm.
answer: 30 cm

## LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

| Output layer | Made | Written from | Verdict |
|---|---|---|---|
| Title<br>the reader-facing framing of the section | 3 | Scope | **grounded** |
| Explanation<br>the principle, rule or relationship to review | 3 | Enacted content | **grounded** |
| Worked example<br>a problem with its solution worked through | 9 | Enacted content | **grounded** |

| | | | |
|---|---|---|---|
| Figure<br>a drawing the learner reads from | 14 | Figures | **grounded** |
| Practice<br>what the learner does | 18 | Assessment<br>written from the teacher's evaluation questions instead of her set work | **substituted** |
| Answer<br>what the learner checks against | 18 | Enacted content | **grounded** |